

# JPA ORM

## New project

New project -> project name: hoten\_mssv\_course\_jpa\_orm

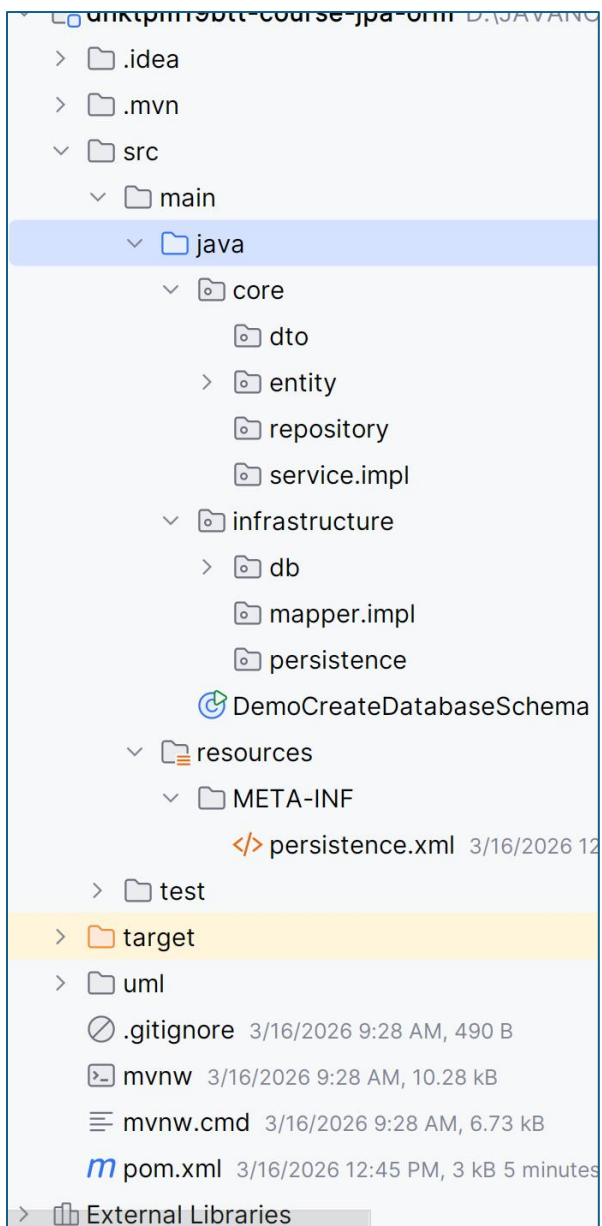
## Add dependencies

dependencies [download](#)

## Persistence.xml

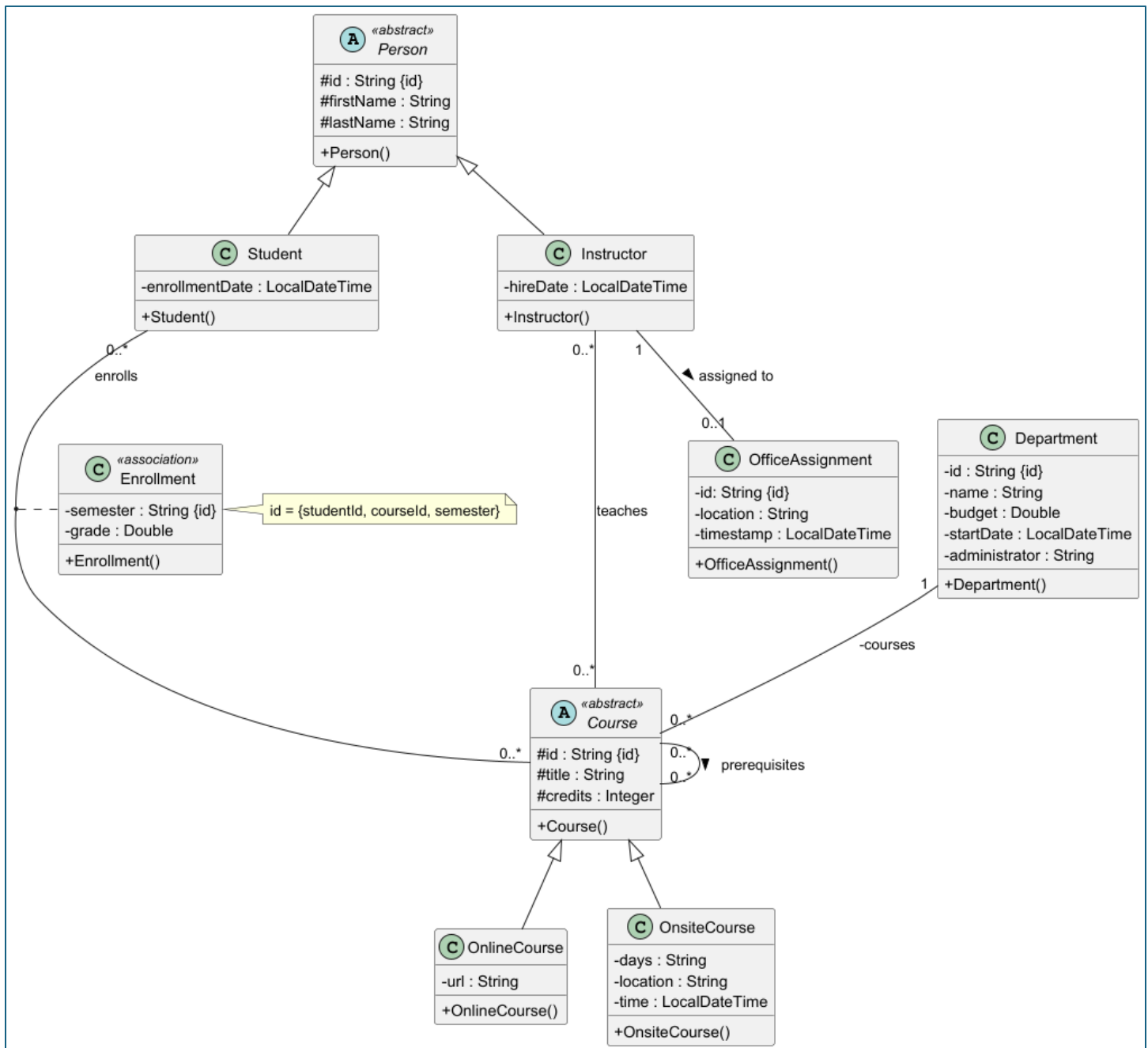
persistence.xml [download](#)

## Project Structure



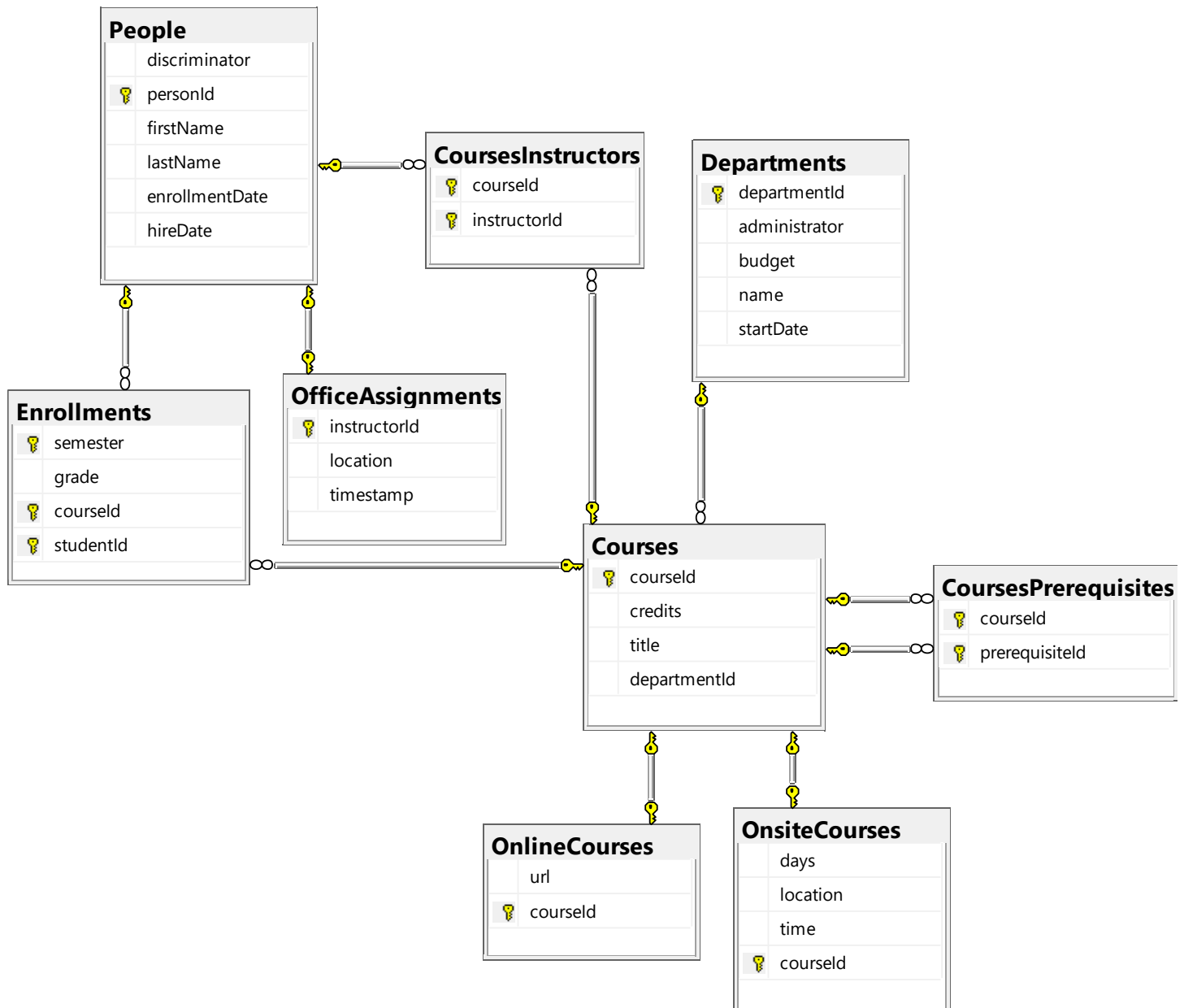
# Class diagram

Puml code [download](#)



Entities class [download](#)

## Database Diagram



## Sample data

Sample data [download](#)

## Query

Using JPQL (*Java Persistence Query Language*), write some methods CRUD to execute the following requests:

1. Create, Update, Delete, Find by Id, Get All on each object.
2. Calculate the number of students in each department, the result is decreasing the number of students.  
+ *getNumberOfStudentsByDepartment ()*: *Map<Department, Long>*
3. Calculate the average score of the students' courses  
+ *getAverageScoreOfStudents()*: *Map<Student, Double>*

4. Departments without students  
+ *listDepartmentsWithoutStudents(): List<Department>*
5. Students studying the subject's name "*Distributed Programming with Java Technology*" have the highest scores.  
+ *listStudentsStudyingCourseWithHighestScore(courseName: String): List<Student>*